8.1 Show and Tell

- Neural Image Caption (NIC) represents a single joint model that takes an image I as input, and is trained to maximize the likelihood $p(S|I)$ of producing a target sequence of words $S = \{S_1, S_2, \ldots\}$, where each word S_t comes from a given dictionary, that describes the image adequately
- the main inspiration comes from machine translation, where the task is to transform a sentence S written in a source language, into its translation T in the target language, by maximizing $p(T|S)$
- this problem is solved by using RNNs
- an encoder RNN reads the source sentence and transforms it into a rich fixed-length vector representation, which in turn is used as the initial hidden state of a decoder RNN that generates the target sentence

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- NIC follows this recipe, replacing the encoder RNN by a deep CNN
- CNNs produce a rich representation of the input image by embedding it to a fixed-length vector, such that this representation can be used for a variety of vision tasks
- hence, it is natural to use a CNN as an image encoder, by first pre-training it for an image classification task and using the last hidden layer as an input to the RNN decoder that generates sentences (see Figure 1)
- NIC is an end-to-end trainable neural network for image captioning
- the model combines sub-networks for vision and language models that can be pre-trained on larger corpora and thus can take advantage of additional data

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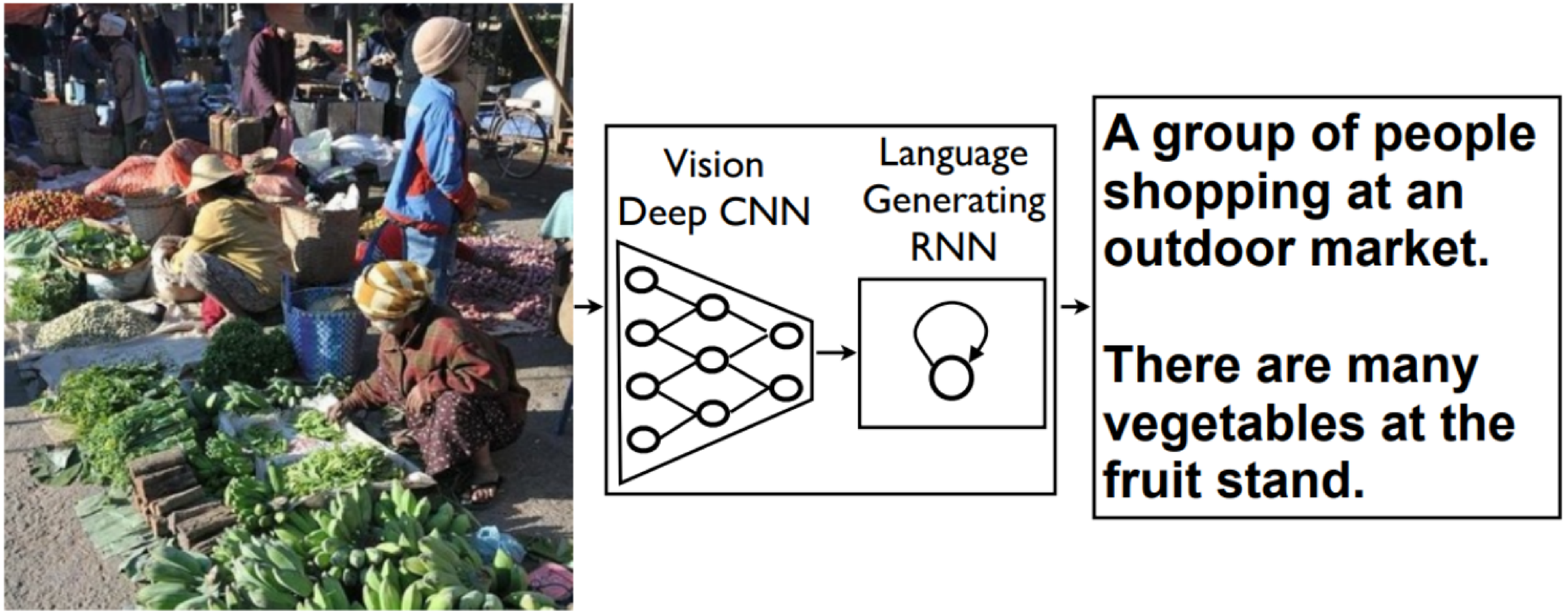


Figure 1: The Neural Image Caption (NIC) model is based end-to-end on a neural network consisting of a vision CNN followed by a language generating RNN. It generates complete sentences in natural language from an input image, as shown in the example.

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- its purpose is to directly maximize the probability $p(S|I)$ of the correct description S given the image I
- since S represents any sentence, its length is unbounded
- thus, it is common to model the joint probability over $S_0, \ldots, S_N$, where N is the length of the description as:

$$p(S|I) = \prod_{t=0}^{N} p(S_t|I, S_0, \ldots, S_{t-1}).$$

- it is natural to model $p(S_t|I, S_0, \ldots, S_{t-1})$ with an RNN, and, for the NIC model, the LSTM network is used
- the LSTM model is trained to predict each word of the sentence after it has seen the image, as well as all preceding words, as defined by $p(S_t|I, S_0, \ldots, S_{t-1})$
- for this purpose, it is instructive to think of the LSTM in unrolled form – a copy of the LSTM memory is created for the image and each sentence word, such that all LSTMs share the same parameters and the output of the LSTM at time $t-1$ is fed to the LSTM at time t (see Figure 2)

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- all recurrent connections are transformed to feed-forward connections in the unrolled version
- in more detail, if we denote by I the input image and by $S = (S_0, \ldots, S_N)$ a true sentence describing this image, the unrolling procedure is as follows:

$$\begin{aligned} x_{-1} &= CNN(I) \\ x_t &= W_e S_t, \quad t \in \{0, \ldots, N-1\} \\ p_{t+1} &= LSTM(x_t), \quad t \in \{0, \ldots, N-1\} \end{aligned}$$

- here, we represent each word as a one-hot vector S_t of dimension equal to the size of the dictionary
- note that we denote by S_0 a special start word and by S_N a special stop word, which designate the start and end of the sentence, respectively
- in particular, by emitting the stop word, the LSTM signals that a complete sentence has been generated
- both the image and the words are mapped to the same space: the image by using a vision CNN, the words by using word embedding matrix W_e

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- the image I is only input once, at $t = -1$, to inform the LSTM about the image contents
- it was empirically verified that feeding the image at each time step as an extra input yields inferior results, as the network can explicitly exploit noise in the image and overfits more easily
- the loss is the sum of the negative log-likelihood of the correct word at each step as follows:

$$L(I, S) = -\sum_{t=1}^{N} \log p_t(S_t).$$

- the above loss is minimized with respect to all the parameters of the LSTM, the top layer of the image embedder CNN, and word embeddings W_e
- all weights were randomly initialized except for the CNN weights, which were pretrained on ImageNet
- the dimension for the embeddings and the size of the LSTM memory were set to 512

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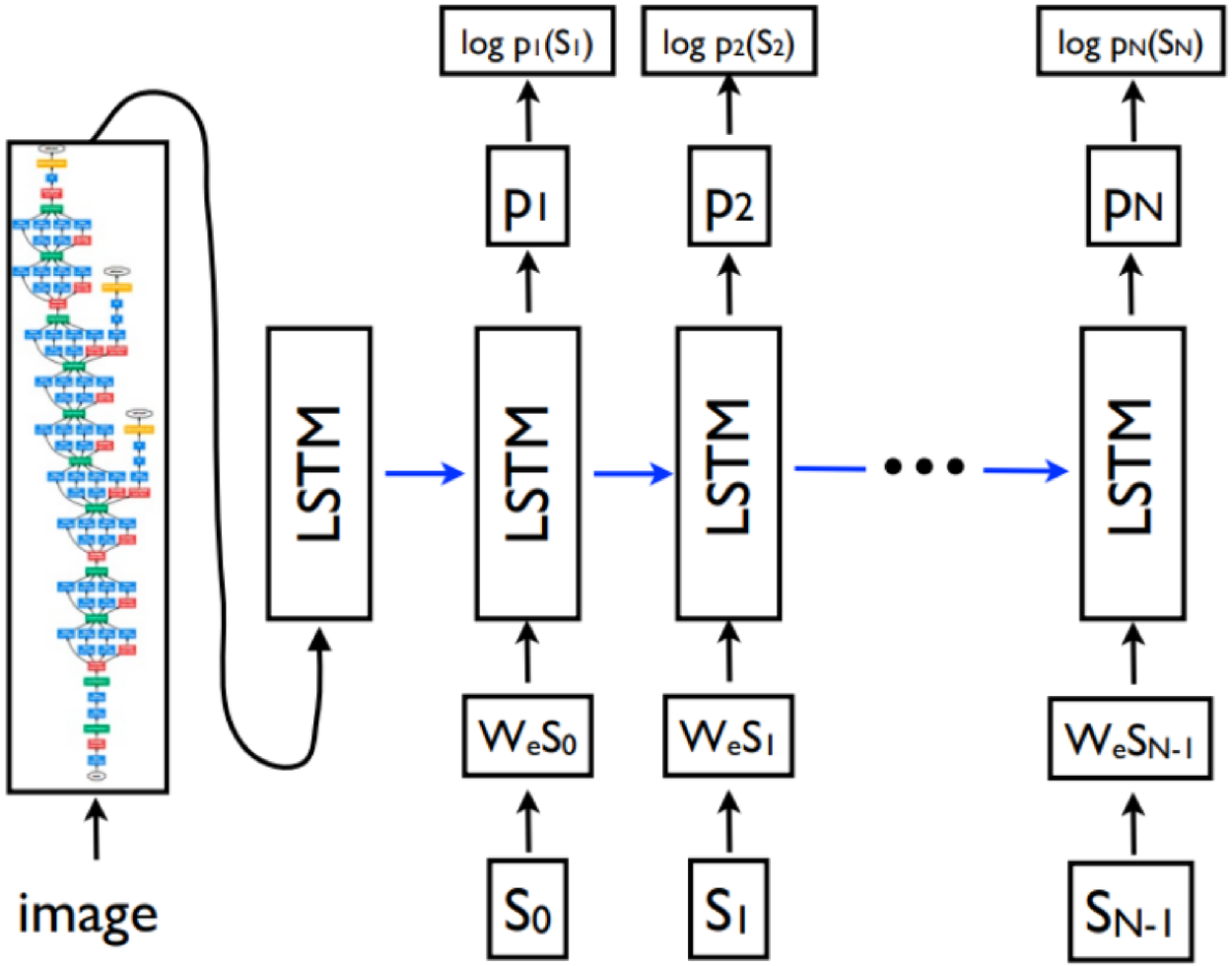


Figure 2: LSTM model combined with a CNN image embedder and word embeddings. The unrolled connections between the LSTM memories are in blue and they correspond to the recurrent connections in the LSTMs. All LSTMs share the same parameters.